

Homework Assignment 3

Instructions. Submit your answers as a PDF on Webcourses by **October 2 at 11:59 p.m.** I encourage you to work in groups and discuss the problems with one another. You may also use AI tools, but remember that AI will not be available during exams. Use these problems to check your own understanding of the material. Your grade is based on participation, not correctness. You must submit your answers to receive credit for this assignment.

Justify every answer. When a relation has a property, prove it from the definition. When it does not have a property, give a specific counterexample. Throughout the assignment, $\mathbb{N} = \{0, 1, 2, \dots\}$.

Question 1 Reading a function from a relation matrix

Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. A relation $R \subseteq A \times B$ has matrix

M_R	1	2	3	4
a	0	1	0	0
b	1	0	0	0
c	0	1	0	0

Rows are indexed by A and columns by B .

- Write R as a set of ordered pairs.
- Does R define a function $f : A \rightarrow B$? Explain using both the ordered pairs and the rows of the matrix. If it does, state its range and determine whether it is one-to-one and whether it is onto.
- For an arbitrary zero-one matrix representing a relation from a finite set X to a finite set Y , state the row condition that makes the relation a function. Assuming it is a function, what column condition characterizes one-to-one functions? What column condition characterizes onto functions?

Question 2 Composition, domains, and inverse behavior

Define

$$f : \mathbb{R} \rightarrow [1, \infty), \quad f(x) = x^2 + 1,$$

and

$$g : [1, \infty) \rightarrow [0, \infty), \quad g(t) = \sqrt{t - 1}.$$

- Compute $(g \circ f)(x)$ for $x \in \mathbb{R}$. Simplify carefully and explain why the answer is $|x|$ rather than always x .
- Compute $(f \circ g)(t)$ for $t \in [1, \infty)$.

- (c) Are f and g inverse functions with the domains and codomains shown above? Explain using both compositions.
- (d) Restrict the domain of f so that f and g become inverse functions. State the restricted function with its domain and codomain, then verify both composition identities.

Question 3 Proving properties of compositions

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions.

- (a) Prove that if $g \circ f$ is one-to-one, then f is one-to-one.
- (b) Prove that if $g \circ f$ is onto C , then g is onto C .
- (c) The converses of these statements are false.

Give explicit finite sets A, B, C and functions $f : A \rightarrow B$ and $g : B \rightarrow C$ showing each of the following:

- (i) f is one-to-one but $g \circ f$ is not one-to-one;
- (ii) g is onto but $g \circ f$ is not onto.

Write each function as a set of ordered pairs and verify the claimed properties.

Question 4 Reflexive, symmetric, and transitive relations

For each relation, determine whether it is reflexive, symmetric, and transitive. Prove every property that holds and give a counterexample for every property that fails.

- (a) Let $U = \{1, 2, 3\}$ and define a relation R on $\mathcal{P}(U)$ by

$$XRY \iff X \subseteq Y.$$

- (b) On $\mathbb{Z}$, define

$$xSy \iff x + y = 0.$$

- (c) Let W be the set of all finite binary strings, including the empty string. Define a relation T on W by

$$uTv \iff u \text{ and } v \text{ have the same length.}$$

For clarity, XRY means $(X, Y) \in R$, and similarly for xSy and uTv .

Question 5 An equivalence relation and its matrix blocks

Let

$$X = \{0, 1, 2\}^2 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)\}.$$

Define a relation $\sim$ on X by

$$(a, b) \sim (c, d) \iff a + b = c + d.$$

Thus $(a, b) \sim (c, d)$ means that the ordered pair $((a, b), (c, d))$ belongs to the relation.

- (a) Prove directly that $\sim$ is reflexive, symmetric, and transitive. Conclude that it is an equivalence relation.
- (b) List every equivalence class. Use each class only once in your list, even though different representatives may name the same class.
- (c) Give an ordering of the nine elements of X in which members of each equivalence class appear consecutively.
- (d) Let J_k denote the $k \times k$ all-ones matrix. Without writing all 81 entries, write the relation matrix in compressed block form after applying your ordering. Explain why the sizes of the J blocks are what they are and why every off-diagonal block is zero.

Question 6 Countability of finite Cartesian powers of $\mathbb{N}$

From lecture, you may use the bijection

$$H : \mathbb{N}^2 \rightarrow \mathbb{N}, \quad H(a, b) = 2^a(2b + 1) - 1.$$

Define $K_3 : \mathbb{N}^3 \rightarrow \mathbb{N}$ by

$$K_3(a, b, c) = H(H(a, b), c).$$

- (a) Compute $K_3(0, 1, 2)$. Then start with the resulting natural number and recover $(0, 1, 2)$ by applying the decoding for H twice.
- (b) Prove that K_3 is one-to-one and onto. You may use the fact that H is a bijection, but make clear where its injectivity and surjectivity are used. Conclude that $\mathbb{N}^3$ is countably infinite.
- (c) Can this construction be generalized to $\mathbb{N}^k$ for every integer $k \geq 1$? Define bijections $K_k : \mathbb{N}^k \rightarrow \mathbb{N}$ recursively and prove by induction on k that each K_k is a bijection.
Hint. Start with $K_1(n) = n$. Given K_k , combine its output with the last coordinate using H to define K_{k+1} .

We begin at $k = 1$ because our convention includes 0 in $\mathbb{N}$; $\mathbb{N}^0$ has one element and is not bijective with $\mathbb{N}$.